

EDUCATION

Southwest Jiaotong University PhD Student, Computer Science	2026.09 – TBD
Advisor: Prof. Dr. Tianrui Li; Can quit the PhD program at any time	
Southwest Jiaotong University M.Sc. Student, Computer Science	2025 – 2026
Advisor: Prof. Dr. Tianrui Li	
Centrale Méditerranée Diplôme d'Ingénieur	2022 – 2025
GPA: 3.57/4.00; 120 ECTS credits; diploma expected after the Jury in October 2026	
Southwest Jiaotong University B.Eng., Electronic Information Engineering	2020 – 2022
Average: 85.19/100; 87.5 Chinese credits	

ACADEMIC RESEARCH

SplAttN: Bridging 2D and 3D with Gaussian Soft Splatting and Attention for Point Cloud Completion	ICML 2026 Spotlight
Zhaoyang Li , Zhichao You, Tianrui Li; Top 2.2%; arXiv: 2605.01466	
- First author with 100% contribution; identified Cross-Modal Entropy Collapse: hard projection maps sparse 3D points to near-empty image support, blocking visual-prior propagation. - Built GS-Bridge: differentiable Gaussian soft splatting converts sparse projections into continuous visual support for geometry-guided 2D–3D feature interaction. - Achieved state-of-the-art on PCN and ShapeNet-55/34; KITTI counterfactuals showed a 26.1% consistency drop after visual removal, validating genuine visual reliance.	
PaCo-VLA: Passivity-Shielded Compliance Prior for Contact-Rich Vision-Language-Action Manipulation	Preprint
Haofan Cao, Zhaoyang Li , Zhichao You, Liang Guo, Tianrui Li; arXiv: 2606.00515	
Contributed to experimental design and manuscript writing for contact-rich Vision-Language-Action manipulation.	

ENGINEERING PROJECTS

CSG Holding Co., Ltd. Large-scale Architectural Glass Defect Inspection System	2025.06 – 2025.11
Live in a real factory environment since 2025.10 ; Python 3.12, OpenCV, FastAPI, WebSocket, Nuitka	
- Designed highly efficient vision algorithms sustaining real-time defect detection from five 2448 x 2048 industrial cameras at >16 FPS on one Intel i5-12400 CPU. - Deployed the line-side system by integrating GigE cameras, multiprocessing acquisition/inference, serial rejection control, and HTTP/WebSocket reporting.	
Arachne Mobile Manipulation Robot	2026.04 – 2026.07
github.com/zay002/Arachne; ROS 2, Scout 2.0, Aubo i5, Gemini335, C16 LiDAR	
- Independently completed hardware, electrical, and software adaptation for Scout 2.0, Aubo i5, gripper, Gemini335 RGB-D, and C16 LiDAR. - Built a ROS 2 integration stack with unified URDF/Xacro and TF, real-robot control, teleoperation, perception, and task workflows.	
VLA Model Adaptation & Deployment π_0 / $\pi_{0.5}$	2025.05 – 2025.07
Target platforms: RM75B and Aubo i5 robotic arms	
- Fine-tuned $\pi_0/\pi_{0.5}$ for RM75B and Aubo i5 by aligning visual observations, joint states, and action spaces. - Deployed the inference stack into ROS 2 for end-to-end robot command execution.	

PATENT

A Method and Related Apparatus for Complex Component Point Cloud Completion Based on Cross-modal Alignment	2026.01
Chinese invention patent application No. 202610232024.2; entered substantive examination	

SKILLS & LANGUAGES

Research: Multimodal Learning, 3D Perception, Robotics, VLA
Engineering: ROS 2, Computer Vision, OpenCV, FastAPI, Python, C++
Languages: Chinese (Native), English (CET-4, CET-6, TOEIC 920/990), French (DELFB2)